

OSAKA INTL, Japan

WMO: 477710

Lat: 34.784N

Lon: 135.439E

Elev: 49

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.9	(c) 31.6	(d) 15.7	(e) 12.2	(f) 37.5	(g) 19.0	(h) 14.4	(i) 38.9	(j) 20.4	(k) 44.6	(l) 18.6	(m) 43.2	(n) 5.4	(o) 320	(p) 0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	142.1	84.1	76.8	139.7	83.9	75.5	133.6	83.5	43.5	90.1	42.6	88.4	41.8	87.3	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.7	(b) 16.5	(c) 14.5	(e) 25.9	(f) 98.5	(g) 1.8	(h) 3.0	(i) 24.6	(j) 100.6	(k) 23.5	(l) 102.4	(m) 22.5	(n) 104.1	(o) 21.2	(p) 106.3		
(a) 18.7	(b) 16.5	(c) 14.5	(e) 23.0	(f) 81.8	(g) 2.0	(h) 0.9	(i) 21.6	(j) 82.4	(k) 20.4	(l) 83.0	(m) 19.3	(n) 83.5	(o) 17.9	(p) 84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	41.0	41.8	47.7	58.7	67.7	74.5	82.2	84.4	77.3	65.9	55.0	45.2
	DBStd	15.86	3.91	4.52	5.40	6.38	4.26	4.21	4.35	3.24	5.28	5.42	5.68	4.67
	HDD50	809	280	232	111	9	0	0	0	0	0	0	16	161
	HDD65	3121	743	648	537	203	23	0	0	0	0	54	301	612
	CDD50	5156	2	4	39	270	548	734	1000	1067	818	494	166	14
	CDD65	1994	0	0	0	14	106	284	535	602	368	83	2	0
	CDH74	19172	0	0	0	64	549	1909	5961	7184	3134	368	3	0
	CDH80	7569	0	0	0	5	71	490	2560	3330	1073	40	0	0
(14) Wind	WSAvg	6.8	6.6	6.5	7.3	7.1	7.1	6.8	7.0	7.3	7.3	6.3	6.0	5.9
(15) Precipitation	PrecAvg	52.8	2.0	2.6	4.0	4.0	5.6	7.0	6.7	4.3	6.4	4.7	3.1	2.3
	PrecMax	68.0	4.1	5.6	6.8	7.1	14.0	15.4	16.3	14.8	15.2	12.0	6.6	4.1
	PrecMin	28.8	0.6	0.8	1.4	1.0	1.3	2.3	1.2	0.8	1.7	1.1	0.7	0.4
	PrecStd	10.5	0.9	1.3	1.3	1.6	3.0	3.0	4.3	3.2	3.3	2.5	1.6	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	63.0	68.4	80.2	85.8	90.0	97.0	98.6	93.4	84.1	73.7	62.8
		MCWB	50.3	53.4	55.3	63.1	66.0	74.4	78.5	78.1	78.1	70.0	62.7	54.4
	2%	DB	53.8	57.1	64.4	75.4	82.1	87.5	94.8	95.2	91.2	80.6	70.2	59.2
		MCWB	46.7	47.7	52.9	60.6	64.9	72.3	78.2	77.9	75.6	68.7	60.8	51.1
	5%	DB	51.7	53.7	61.1	73.1	79.1	84.6	92.8	93.4	88.2	78.4	68.0	57.2
		MCWB	45.3	46.1	52.2	59.2	64.0	71.0	77.6	77.4	74.7	66.5	59.6	49.9
	10%	DB	49.7	51.4	58.7	69.9	77.1	82.6	90.0	91.6	86.1	75.4	64.8	55.0
		MCWB	43.5	44.8	50.1	59.2	63.5	70.4	77.0	77.2	74.0	65.5	57.5	48.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	54.5	59.3	68.8	72.8	78.1	80.8	81.0	79.7	74.3	65.7	57.3
		MCDB	56.8	60.5	63.7	73.9	77.9	85.5	91.4	90.5	88.4	78.7	69.2	60.6
	2%	WB	49.0	50.4	56.3	65.5	70.2	76.5	79.8	80.1	78.3	71.4	63.4	53.5
		MCDB	51.7	54.2	61.7	70.4	74.6	82.6	90.2	89.8	86.4	77.8	68.3	57.6
	5%	WB	45.9	48.0	54.0	62.8	68.0	75.2	78.9	79.5	77.3	69.1	61.3	50.4
		MCDB	50.5	52.5	60.1	69.2	73.8	81.1	88.7	88.8	85.1	75.4	66.3	55.5
	10%	WB	43.2	44.9	51.2	60.2	65.9	73.5	78.1	78.7	76.1	66.9	59.2	48.2
		MCDB	48.4	50.3	56.7	67.4	73.9	79.5	87.2	87.9	83.3	73.8	64.5	54.2
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.3	16.5	17.8	17.3	14.2	13.5	14.9	14.3	16.0	16.5	15.4
		MCDBR	17.3	20.1	20.9	21.3	20.8	17.7	16.4	17.0	16.5	17.9	17.7	17.9
		MCWBR	12.5	13.5	13.7	10.9	9.8	7.8	5.7	5.8	6.6	8.6	11.3	11.7
	5% WB	MCDBR	15.2	16.1	16.9	15.8	14.2	12.3	14.4	14.5	14.1	14.2	15.3	15.8
		MCWBR	12.7	13.0	13.6	11.2	9.9	6.6	5.6	5.6	6.5	8.7	11.4	11.9
(40) Clear-Sky Solar Irradiance	taub		0.417	0.461	0.525	0.558	0.563	0.584	0.585	0.551	0.503	0.461	0.432	0.412
	taud		2.090	1.994	1.855	1.829	1.880	1.898	1.945	2.023	2.125	2.156	2.137	2.121
	Ebn at Noon		239	242	237	237	238	232	231	236	241	240	232	231
	Edh at Noon		40	49	62	66	63	62	59	54	46	42	38	37
(44) All-Sky Solar Radiation	RadAvg		686	889	1212	1495	1644	1491	1574	1614	1268	1022	765	623
	RadStd		49	74	97	131	168	139	211	149	107	99	62	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page